

MATH 1203 CIS & BIT DISCRETE MATHEMATICS

Lecture 2: COUNTING TECHNIQUES

Course Lecturer: M. KANYIKA

Applied Studies Department- MUST

FEBRUARY, 2026

Lesson Outcome

By the end of this lesson you should be able to:

- Define Sum rule as a counting techniques
- Demonstrate understanding and ability to solve problems using sum rule technique.
- Describe how Sum rule can be applied in solving one-to-one functions
- Combine Product rule and sum rule to solve more complex problems

INTRODUCTION

- I present to you the second lecture of this course on the topic, Counting techniques.
- In the previous lecture, you learnt about the product rule as counting technique
- In this lecture, you will be introduced to other counting technique.
- We will Look on the Sum rule as counting techniques
- how we can combine both sum and product rule to solve complex problems

The Sum Rule

Definition 1

Suppose some task A can be done in m ways and a second task B can be done in n ways, and suppose that both tasks cannot be done simultaneously, then by sum rule there are $m + n$ ways of doing a task A or B .

The following examples illustrate how sum rule is used.

Sum Rule CONT'D

Example 2

Suppose a College has 3 different history courses, 4 different literature courses and 2 different sociology courses. How many ways can a student choose just one of the courses?

Solution: The number n of ways a student can choose just one of the courses is:

$$\begin{aligned} n &= m_1 + m_2 + m_3 \\ &= 3 + 4 + 2 \\ &= 9 \end{aligned}$$

Sum Rule CONT'D

Example 3

A student can choose a computer project from one of the three lists. The lists contains 23 ,15 and 19 possible projects , respectively. No project is more than one list. How many possible projects are there to choose from?

Solution

- The student can choose a project by selecting a project from the first list, second list and third list.

Sum Rule CONT'D

- Because no project is on more than one list,
- By sum rule there are $23+15+19=57$ ways to choose a project.

The Sum Rule in terms of Sets

- The sum rule can also be phrased in terms of sets.
- If $A_1, A_2, \dots, A_m$ are disjoint finite sets, then the number of elements in the union of these sets is the sum of the numbers of elements in the sets.
- This implies that there are $|A_i|$ ways to choose an element from $|A_i|$ for $i = 1, 2, \dots, m$.

Sum Rule in terms of sets CONT'D

- By the sum rule , the number of ways to choose an element from one of the sets, which is the number of the elements in the union is given as follows:

$$| A_1 \cup A_2 \cup \dots \cup A_m | = | A_1 | + | A_2 | + | A_3 | \dots + | A_m |$$

Combining Sum Rule and Product Rule

Combining the sum and product rule allows us to solve more complex problems.

Example 4

Suppose statement labels in a programming language can be either a single letter or a letter followed by a digit. Find the number of possible labels.

Solution

- Single upper case letter = 26
- Upper case letter **followed** by a digit = 26.10

Product and Sum Rule CONT'D

- Single upper case letter **or** Upper case letter **followed** by a digit
- This implies combination of product and sum rule.
- Hence there are $26 + 26 \cdot 10 = 286$ possible labels.

Counting Passwords

Example 5

Each user on a computer system has a password, which is six to eight characters long, where each character is an uppercase letter or a digit. Each password must contain at least one digit. How many possible passwords are there?

Solution

- Let P be the total number of passwords, and let P_6 , P_7 , and P_8 be the passwords of length 6, 7, and 8.
- By the sum rule $P = P_6 + P_7 + P_8$.

Counting Passwords solution CONT'D

- To find each of P_6 , P_7 , and P_8
- We find the number of passwords of the specified length composed of letters and digits and subtract the number composed only of letters
- We find that:

$$\begin{aligned}P_6 &= (\text{letters} + \text{digits})^6 - (\text{letters})^6 \\&= 36^6 - 26^6 \\&= 2,176,782,336 - 308,915,776 \\&= 1,867,866,560.\end{aligned}$$

Counting Passwords solution CONT'D

- Similarly , it can also be shown that

$$\begin{aligned}P_7 &= (\text{letters} + \text{digits})^7 - (\text{letters})^7 \\&= 36^7 - 26^7 \\&= 78,364,164,096 - 8,031,810,176 \\&= 70,332,353,920.\end{aligned}$$

Counting Passwords solution CONT'D

- And , it can also be shown that

$$\begin{aligned}P_8 &= (\text{letters} + \text{digits})^8 - (\text{letters})^8 \\&= 36^8 - 26^8 \\&= 2,821,109,907,456 - 208,827,064,576 \\&= 2,612,282,842,880.\end{aligned}$$

Therefore, $P = P_6 + P_7 + P_8 = 2,684,483,063,360$.

Conclusion

- In this lesson, you have learnt about the following:
 - (a) Product rule and Sum Rule
 - (b) combining product and sum rule
 - (c) Counting Passwords
- If you have any questions about this lecture, please do not hesitate to contact us using the discussion forum on this platform and we will be happy to assist you.
- I am convinced that this lecture was well presented and you have understood it.
- I wish you all the best.